



SALEDUCATION
SALENGINEERINGANDTECHNICALINSTITUTE

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Student's Weekly Report of Internship

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Enrollment Number:	201260107012		
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Work done in last Week with description :

Week 1 (Date : 9th February 2024 to 15th February 2024) :

1) Tools are use by CID

Plans for next week:

1) Basic Knowlegde About anti-bullying unit

Task

- Tools are use by CID

- Forensic Science Laboratories (FSL): FSLs play a crucial role in criminal investigations by analyzing physical evidence such as fingerprints, DNA samples, bloodstains, firearms, and other trace evidence. They use advanced techniques and equipment to identify suspects and provide scientific evidence for prosecution.
 - Microscopes: Microscopes are essential tools used to examine various types of evidence, such as hair, fibers, soil, and gunshot residue. They help forensic scientists analyze the characteristics and properties of microscopic particles to identify potential matches or associations with suspects or crime scenes.
 - DNA Analysis Equipment: FSLs are equipped with sophisticated DNA analysis equipment, including PCR (Polymerase Chain Reaction) machines, DNA sequencers, and genetic analyzers. These tools enable forensic scientists to extract, amplify, and analyze DNA samples from biological evidence like blood, saliva, hair, and tissue to identify individuals or establish familial relationships.
 - Fingerprinting Tools: Fingerprint analysis is a fundamental aspect of forensic investigations. FSLs utilize various tools such as fingerprint powders, brushes, lifting tape, and chemical reagents to develop, enhance, and collect latent fingerprints from surfaces. Automated fingerprint identification systems (AFIS) are also used to match and compare fingerprints stored in databases.
 - Ballistics and Firearms Examination Tools: FSLs employ specialized tools for the analysis of firearms, ammunition, and ballistic evidence. This includes comparison microscopes, bullet trajectory analysis kits, and gunshot residue detection kits to examine bullet casings, bullets, and gunpowder residues for identification and analysis.
 - Toxicology Instruments: Toxicology laboratories within FSLs are equipped with instruments like gas chromatographs, mass spectrometers, and immunoassay analyzers to detect and quantify drugs, alcohol, poisons, and other toxic substances in biological samples such as blood, urine, and tissue.
 - Digital Forensics Software: With the proliferation of digital devices, FSLs utilize specialized software tools for digital forensics to extract, analyze, and recover data from computers, mobile phones, hard drives, and other digital storage media. These tools help forensic examiners retrieve deleted files, analyze internet browsing history, and recover encrypted data for criminal investigations.

- Chemical Analysis Equipment: FSLs have a wide range of chemical analysis equipment, including spectrophotometers, chromatography systems, and spectroscopy instruments, to analyze substances such as drugs, explosives, accelerants, and chemical residues found at crime scenes.
 - Evidence Packaging and Preservation Supplies: FSLs use specialized packaging materials such as evidence bags, containers, seals, and labels to properly collect, package, and preserve physical evidence for analysis while maintaining chain of custody.
- Cyber Forensics Tools: With the rise of cybercrime, CID employs specialized tools for digital forensics to gather evidence from computers, mobile devices, and digital storage media. These tools help in recovering deleted data, analyzing malware, tracing online activities, and identifying cybercriminals.
- EnCase Forensic: EnCase Forensic is a widely used digital investigation platform that allows forensic examiners to collect, preserve, and analyze digital evidence from various sources, including computers, mobile devices, and cloud storage. It provides features for disk imaging, file recovery, keyword searching, and timeline analysis.
 - AccessData Forensic Toolkit (FTK): FTK is a comprehensive forensic analysis tool that assists investigators in acquiring, analyzing, and presenting digital evidence. It supports forensic imaging, data carving, keyword searching, email analysis, and artifact extraction from various operating systems and file systems.
 - Autopsy: Autopsy is an open-source digital forensics platform used for analyzing disk images and data recovery. It offers a user-friendly interface and features for timeline analysis, keyword searching, file carving, and registry analysis. Autopsy is widely used by law enforcement agencies, government organizations, and forensic professionals.
 - Volatility Framework: Volatility is a powerful memory forensics framework used to extract and analyze volatile memory (RAM) from live systems. It helps investigators identify running processes, open network connections, loaded drivers, and other artifacts that may be crucial for understanding system activity and identifying malware.
 - Wireshark: Wireshark is a popular network protocol analyzer used for capturing and analyzing network traffic in real-time. It allows forensic examiners to inspect packets, reconstruct network sessions, and identify suspicious activities or security breaches. Wireshark supports a wide range of protocols and can be used for network forensics investigations.
 - Sleuth Kit: Sleuth Kit is an open-source toolkit for digital forensic analysis of disk images and file systems. It provides command-line tools for file system analysis, file extraction, metadata examination, and timeline creation. Sleuth Kit supports various file systems, including NTFS, FAT, EXT, and HFS+.

- Cellebrite UFED (Universal Forensic Extraction Device): UFED is a forensic tool used for extracting and analyzing data from mobile devices, including smartphones, tablets, and GPS devices. It supports physical and logical acquisitions, password bypassing, data decoding, and analysis of device backups and cloud storage.
 - XRY: XRY is another popular mobile forensic tool used to acquire and analyze data from mobile devices and SIM cards. It supports a wide range of devices, including smartphones, feature phones, and GPS devices. XRY can extract call logs, messages, contacts, images, and other data from mobile devices for forensic analysis.
- Crime Scene Investigation (CSI) Kits: CID teams are equipped with CSI kits containing essential tools and supplies for collecting and preserving physical evidence at crime scenes. These kits typically include items such as fingerprinting materials, evidence bags, swabs, gloves, and measuring tools.
- Fingerprinting Supplies: Fingerprinting is a crucial aspect of forensic investigation. CSI kits include materials such as fingerprint powders, brushes, lifting tape, and fingerprint cards. These tools are used to develop, enhance, and collect latent fingerprints from various surfaces.
 - Evidence Collection Tools: CSI kits contain a variety of tools for collecting and packaging different types of evidence. This may include evidence bags, containers, envelopes, swabs, tweezers, and gloves. These tools help investigators collect biological samples, hair, fibers, weapons, shell casings, and other physical evidence without contaminating or damaging it.
 - Photography Equipment: Documentation of the crime scene is essential for reconstructing events and preserving evidence. CSI kits often include cameras, tripods, and rulers for capturing detailed photographs of the scene, including overall views, close-ups of evidence, and measurements for scale.
 - Evidence Markers and Labels: To accurately document the location of evidence within the crime scene, CSI kits provide markers, flags, cones, and labels. These items help investigators mark and label evidence, document its position relative to other objects, and maintain the chain of custody.
 - Personal Protective Equipment (PPE): Safety is paramount at crime scenes to prevent contamination and ensure the well-being of investigators. CSI kits typically include PPE such as gloves, masks, goggles, and shoe covers to protect investigators from biological hazards, chemical exposure, and other risks.
 - Lighting and Magnification Tools: Proper illumination and magnification are crucial for detecting and examining trace evidence. CSI kits may include flashlights, UV lights, light sources, and magnifying lenses to enhance visibility and reveal hidden details such as bloodstains, fibers, and gunshot residue.
 - Evidence Documentation Forms: To maintain accurate records of the crime scene investigation, CSI kits provide documentation forms and logbooks for recording

observations, sketches, measurements, and descriptions of evidence. These forms ensure that investigators document their findings systematically and comprehensively.

- **Barrier Tape and Crime Scene Tape:** Establishing a secure perimeter around the crime scene is essential to preserve evidence and prevent contamination. CSI kits may include barrier tape, crime scene tape, and signage to cordon off the area and restrict access to authorized personnel only.
- **Surveillance Equipment:** CID may use surveillance tools such as cameras, audio recording devices, GPS trackers, and drones to monitor suspects, gather intelligence, and gather evidence covertly.
 - **Closed-Circuit Television (CCTV) Cameras:** CCTV cameras are widely used for video surveillance in public spaces, businesses, homes, and government facilities. These cameras capture live video footage and can be fixed, pan-tilt-zoom (PTZ), or dome cameras, depending on the application. CCTV systems often include DVRs (Digital Video Recorders) or NVRs (Network Video Recorders) for recording and storing video footage.
 - **Hidden Cameras:** Also known as covert cameras or spy cameras, hidden cameras are designed to be discreet and blend into their surroundings. They can be disguised as everyday objects such as clocks, smoke detectors, pens, or picture frames. Hidden cameras are commonly used for surveillance in undercover operations, sting operations, or to monitor activities without detection.
 - **Audio Surveillance Devices:** Audio surveillance devices, such as microphones, voice recorders, and bug detectors, are used to capture conversations and sounds covertly. These devices may be installed in rooms, vehicles, or on individuals to gather evidence or intelligence. However, the use of audio surveillance equipment is subject to legal restrictions in many jurisdictions.
 - **GPS Trackers:** GPS (Global Positioning System) trackers are used to monitor the location and movements of vehicles, assets, or individuals in real-time. These devices use satellite signals to determine precise location coordinates, which can be tracked remotely using software or mobile apps. GPS trackers are commonly used for fleet management, asset tracking, and personal safety.
 - **Drones:** Drones, also known as unmanned aerial vehicles (UAVs), are equipped with cameras and sensors for aerial surveillance and reconnaissance. Drones can capture high-definition video footage and images from various vantage points, making them valuable tools for surveillance of large areas, crowd monitoring, and search-and-rescue operations.
 - **Body-Worn Cameras:** Body-worn cameras are small, portable cameras worn by law enforcement officers, security personnel, or individuals to record interactions, incidents, or events from a first-person perspective. Body-worn cameras are used to enhance transparency, accountability, and evidence

collection during law enforcement operations, security patrols, or personal safety situations.

- **Wireless Video Transmitters and Receivers:** Wireless video transmitters and receivers are used to transmit live video and audio signals wirelessly over short or long distances. These devices enable remote monitoring and surveillance in situations where running cables is impractical or not feasible, such as outdoor surveillance or covert operations.
 - **Facial Recognition Systems:** Facial recognition systems use biometric technology to identify or verify individuals based on their facial features. These systems analyze facial images captured by surveillance cameras or other sources and compare them against databases of known faces to identify or track individuals of interest.
- **Communication and Interrogation Tools:** CID utilizes communication tools such as telephones, radios, and encrypted messaging systems to coordinate investigations and share information securely. Interrogation tools may include recording devices, polygraph machines, and psychological profiling techniques to elicit information from suspects and witnesses.
- **Telephones:** CID investigators use telephones for communication purposes, including making and receiving calls to gather information, coordinate operations, and communicate with colleagues, witnesses, informants, and other law enforcement agencies.
 - **Radios:** Two-way radios are utilized by CID personnel for instant communication during field operations, surveillance, and coordination with other law enforcement units. Radios provide a secure and efficient means of communication, especially in areas where cellular networks may be unavailable or unreliable.
 - **Recording Devices:** CID investigators use digital voice recorders, video cameras, and body-worn cameras to record interviews, interrogations, statements, and interactions with suspects, witnesses, and informants. Recording devices ensure accurate documentation of proceedings and preserve evidence for legal purposes.
 - **Interrogation Rooms:** CID maintains specialized interrogation rooms equipped with audio and video recording equipment, one-way mirrors, observation rooms, and other facilities to conduct formal interviews and interrogations with suspects and witnesses. Interrogation rooms provide a controlled environment for questioning while ensuring transparency and accountability in the investigative process.
 - **Polygraph Machines:** CID may use polygraph machines during interrogations to assess the credibility of suspects and witnesses by measuring physiological responses such as heart rate, blood pressure, and respiration. Polygraph examinations can help investigators identify deception and guide further questioning.
 - **Communication Analysis Software:** CID employs communication analysis software to analyze and visualize communication data, including phone records, text messages, emails, and social media interactions. These tools help investigators identify patterns,

connections, and associations among individuals and entities involved in criminal activities.

- **Psychological Profiling Techniques:** CID investigators utilize psychological profiling techniques, such as behavioral analysis and criminal profiling, to understand the behavior, motives, and characteristics of suspects and offenders. Psychological profiling aids in developing strategies for interviewing and interrogating suspects effectively and eliciting valuable information.
 - **Translator Services:** In cases involving non-English speaking individuals or foreign languages, CID may utilize translator services to facilitate communication and interviews with suspects, witnesses, and victims. Translators help ensure accurate interpretation of statements and facilitate information exchange during investigations.
- **Geospatial Technology:** Geographic Information Systems (GIS) and mapping software are used by CID to analyze crime patterns, visualize spatial data, and identify hotspots for criminal activity. This helps in strategic planning, resource allocation, and crime prevention initiatives.
- **Crime Mapping and Analysis:** Geospatial technology allows CID to create crime maps by plotting the locations of crimes such as thefts, burglaries, assaults, and homicides on digital maps. By visualizing crime patterns and hotspots, CID can identify trends, allocate resources more effectively, and develop targeted crime prevention strategies.
 - **Spatial Analysis:** CID uses spatial analysis techniques to analyze the spatial relationships between crime incidents, demographics, socioeconomic factors, and environmental features. This helps investigators identify underlying causes and contributing factors to crime, such as poverty, unemployment, or lack of street lighting.
 - **Geographic Information Systems (GIS):** GIS software enables CID to collect, manage, analyze, and visualize geographic data related to crime incidents, suspects, witnesses, and other relevant information. CID can overlay crime data with demographic data, land use information, and other spatial datasets to gain insights into crime patterns and dynamics.
 - **Crime Scene Mapping:** Geospatial technology allows CID to create detailed maps of crime scenes, including the locations of evidence, witness statements, and other relevant information. Crime scene mapping helps investigators reconstruct events, identify potential suspects, and present evidence more effectively in court.
 - **Suspect Tracking and Surveillance:** Geospatial technology can be used to track the movements of suspects and persons of interest using GPS tracking devices or mobile phone tracking. By visualizing the locations visited by suspects over time, CID can gather evidence of their activities and connections to criminal incidents.

- **Field Operations Planning:** CID utilizes geospatial technology for planning and coordinating field operations, such as surveillance, raids, and search operations. GIS tools help CID identify target areas, plan routes, allocate resources, and assess operational risks based on geographical factors such as terrain, infrastructure, and population density.
 - **Disaster Management and Emergency Response:** In addition to crime investigation, CID may use geospatial technology for disaster management and emergency response. GIS helps CID identify vulnerable areas, assess risks, plan evacuation routes, and coordinate response efforts during natural disasters, man-made emergencies, or public safety incidents.
 - **Public Safety and Community Policing:** Geospatial technology supports CID's efforts in public safety and community policing initiatives. By analyzing spatial data on crime, demographics, and community resources, CID can develop crime prevention programs, engage with communities, and target interventions in high-risk areas to improve public safety and build trust with the community.
- **Database and Records Management Systems:** CID maintains databases and records management systems to store and retrieve information related to criminal cases, suspects, witnesses, and past criminal activities. These systems enable efficient data management, analysis, and information sharing among law enforcement agencies.
- **Case Management Systems:** CID employs case management systems to track and manage information related to ongoing criminal cases. These systems allow investigators to create case files, record case details, log evidence, document investigative steps, and track case progress from initiation to resolution.
 - **Suspect and Witness Databases:** CID maintains databases of suspects, persons of interest, and witnesses associated with criminal investigations. These databases store information such as personal details, physical descriptions, criminal histories, known associates, and contact information to aid in suspect identification and witness coordination.
 - **Evidence Management Systems:** CID utilizes evidence management systems to catalog, track, and manage physical evidence collected during criminal investigations. These systems maintain detailed records of each piece of evidence, including its description, chain of custody, storage location, and analysis status, to ensure its integrity and admissibility in court.
 - **Fingerprint and DNA Databases:** CID may have access to fingerprint and DNA databases, such as Automated Fingerprint Identification Systems (AFIS) and Combined DNA Index Systems (CODIS), to compare and match fingerprints and DNA profiles collected from crime scenes, suspects, and convicted offenders to aid in criminal identification and investigation.
 - **Crime Data Analysis Systems:** CID employs crime data analysis systems to analyze and visualize crime data, including patterns, trends, and hotspots. These systems integrate data from various sources, such as crime reports, incident logs, and dispatch

records, to identify crime patterns, allocate resources, and inform strategic decision-making.

- **Intelligence and Information Sharing Platforms:** CID collaborates with other law enforcement agencies and stakeholders through intelligence and information sharing platforms. These platforms facilitate the exchange of intelligence, case-related information, and investigative leads among CID units, local police departments, state agencies, federal authorities, and international partners to enhance coordination and cooperation in combating crime.
 - **Digital Forensics Databases:** CID maintains databases of digital evidence collected during digital forensics investigations, including computer files, emails, internet activity logs, and social media data. These databases support digital evidence analysis, preservation, and retrieval to uncover digital traces and electronic evidence relevant to criminal investigations.
 - **Legal and Administrative Records:** CID manages legal and administrative records, such as arrest records, court documents, warrants, subpoenas, and administrative correspondence, related to criminal investigations and judicial proceedings. These records ensure compliance with legal requirements, support prosecution efforts, and provide documentation of case-related activities.
- **Collaborative Platforms:** CID may utilize collaborative platforms and case management software to facilitate teamwork, share updates, and coordinate efforts among investigators, forensic experts, prosecutors, and other stakeholders involved in complex cases.
- **Secure Messaging Systems:** CID employs secure messaging systems to exchange sensitive information, updates, and investigative leads securely among investigators, law enforcement agencies, and partner organizations. These platforms ensure confidentiality and data integrity while facilitating real-time communication and collaboration.
 - **Case Management Software:** CID utilizes case management software to centralize case-related information, documentation, and evidence in a secure and organized manner. These platforms enable investigators to create, update, and track case files, assign tasks, manage deadlines, and collaborate on case-related activities efficiently.
 - **Investigative Collaboration Portals:** CID may develop or utilize investigative collaboration portals to provide investigators with access to centralized repositories of case-related resources, training materials, best practices, and analytical tools. These portals serve as hubs for knowledge sharing, collaboration, and professional development within the investigative community.
 - **Digital Evidence Sharing Platforms:** CID leverages digital evidence sharing platforms to facilitate the exchange of digital evidence, forensic findings, and investigative insights with forensic laboratories, digital forensic examiners, and other relevant stakeholders. These platforms streamline the sharing and analysis of digital evidence, enhancing the efficiency and effectiveness of investigations.

- **Intelligence Fusion Centers:** CID collaborates with intelligence fusion centers to gather, analyze, and disseminate intelligence on emerging threats, criminal trends, and high-priority cases. These centers serve as hubs for multi-agency collaboration, information fusion, and strategic coordination to enhance situational awareness and support proactive law enforcement efforts.
- **Task Force Collaboration Tools:** CID participates in task forces and joint investigative teams focused on specific types of crime or criminal activities. Task force collaboration tools, such as shared calendars, document repositories, and virtual workspaces, facilitate communication, task assignment, and information sharing among task force members from different agencies and jurisdictions.
- **Secure File Sharing Platforms:** CID utilizes secure file sharing platforms to exchange large files, documents, and multimedia evidence securely with internal and external stakeholders. These platforms provide encryption, access controls, and audit trails to ensure the confidentiality and integrity of shared files during transit and storage.
- **Virtual Meeting and Conferencing Tools:** CID employs virtual meeting and conferencing tools to conduct remote meetings, briefings, and case conferences with investigators, experts, and stakeholders located in different geographic locations. These tools enable real-time collaboration, information sharing, and decision-making without the need for physical presence.

Total Working Hrs: —~~25 Hours~~—

Signature of Student:

The above entries are correct and the grading of work done by Trainee is

Excellent	Very Good	Good	Fair	Below Average	Poor

Signature of External Guide with Company Seal:

Signature of Internal Guide

Date:

Date: